Mechanical Mechanisms

Lifting Machine

			
Mohamed Elsherbini	Mohammed Taha	Abdullah Aqle	Mohammad Kamrul
202307841	202309234	202309898	202313375

Instructor: Prof. Dr. Asan Abdul Muthalif

Lab instructor: Eng. Hammam Anazeh

Qatar University

Fall 2025

Department of Mechanical and Industrial Engineering Course: MECH-321 Mechanical Mechanisms

Section: L02

Submission date: 05-Dec-2025

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Note: Each links combinations result in different trace path graphs made by the two hooks transferring the object from the ground to the higher level.

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Abstract:
This project focuses on the design, modeling, and analysis of a mechanical lifting system inspired by the perpetual lifting machine demonstrated in the provided reference video. The objective was to develop a fully functional mechanism driven by a single hand-operated crank and supported by a gear-based linkage system to produce a continuous lifting motion. The work includes conceptual sketching, kinematic synthesis using Linkage software, detailed 3D CAD modeling in SolidWorks, and motion analysis to obtain position, velocity, acceleration, and dynamic force results. A pair of gears was integrated into the design as a constraint requirement to ensure proper transmission and synchronization of motion throughout the lifting cycle. The final virtual prototype was animated using a constant angular velocity input, and the resulting kinematic plots and joint forces were extracted for evaluation. Additionally, analytical equations for displacement, velocity, acceleration, and driving torque were developed to support the computational results. A physical prototype was constructed to validate the mechanism's performance and compare real-world behavior with the simulated outcomes. The project demonstrates a complete workflow of mechanism design, kinematic analysis, dynamic loading evaluation, and prototype realization.

1. Introduction

1.1 Problem Statement/Identification

Mechanical lifting systems are widely used in industrial and domestic applications, often requiring synchronized motion, continuous lifting action, and efficient power transmission. The problem addressed in this project is the design and analysis of a mechanism capable of raising an object through a repeated lifting cycle driven by a single rotary input. The mechanism must emulate the motion demonstrated in the reference perpetual lifting machine, where two sides alternately carry the load throughout the cycle. Additionally, the system must incorporate a gear, a cam, a belt, or a chain, while maintaining smooth and continuous motion. The challenge lies in translating the visual concept into a functional linkage mechanism, achieving correct kinematic behavior, and validating its performance through simulation and physical prototyping.

1.2 Motivation
Understanding how linkage mechanisms operate is an essential part of mechanical engineering, especially in applications involving automation, materials handling, and motion control. This project offers the opportunity to combine theoretical knowledge with practical design skills by converting an observed mechanism into a complete working system. The motivation behind the work is to build competence in kinematic synthesis, CAD modeling, motion simulation, and gear selection, which are all important skills for real engineering applications. Constructing a physical prototype also strengthens practical understanding and allows direct comparison between theoretical predictions and actual performance.

1.3 Project Objectives

The objectives of this project are as follows:
1. To study the reference video and create conceptual sketches that identify the main links and joints of the mechanism.

Note: Each links combinations result in different trace path graphs made by the two hooks transferring the object from the ground to the higher level.

2. To model and refine the mechanism using Linkage software and then create a full 3D CAD model in SolidWorks with a crank-driven input.
3. To select and integrate a gear as the required motion transmission component and ensure that it operates correctly within the mechanism.
4. To perform position, velocity, and acceleration analysis using SolidWorks Motion and generate plots of the kinematic variables against the input crank angle.
5. To determine joint forces and the driving torque needed to maintain constant angular velocity of the input crank.
6. To develop analytical equations for the position, velocity, acceleration, and force behavior of the mechanism.
7. To construct a physical prototype and compare its performance to the virtual model.

1.4 Work Distribution

https://www.vertex42.com/ExcelTemplates/simple-gantt-chart.html

TASK	ASSIGNED TO	PROGRESS	START	END
Initiation				
Define Project Objectives	Abdalla Aqle , El Sherbini , Taha , Kamrul	100%	9/25/25	9/25/25
Brainstorm Mechanism Ideas	Mohamed El Sherbini , Aqle , Taha , Kamrul	100%	9/26/25	9/26/25
Sketching	Mohamed El Sherbini , Aqle , Taha , Kamrul	100%	9/26/25	9/28/25
Select Mechanism Type	Mohamed El Sherbini , Aqle , Taha , Kamrul	100%	9/28/25	9/28/25
Identify Key Components	Mohammed Taha , Aqle, Kamrul , El Sherbini	100%	9/28/25	9/30/25
Execution				
Solidworks Modelling	Mohammed Taha , Aqle, Kamrul , El Sherbini	100%	9/29/25	10/3/25
Mechanism Assembly	Mohammed Taha , Aqle, Kamrul , El Sherbini	100%	10/1/25	10/6/25
Animation	Abdalla Aqle , El Sherbini , Taha , Kamrul	100%	10/6/25	10/9/25
Analytical Calculations	Mohammad Kamrul, Taha , El Sherbini , Aqle	100%	10/6/25	10/18/25
Comparision and Validation	Mohammad Kamrul, Taha , El Sherbini , Aqle	100%	10/6/25	10/20/25
Adjusting Lengths	Mohamed El Sherbini , Aqle , Taha , Kamrul	100%	10/7/25	10/21/25
Constraints Analysis	Abdalla Aqle , El Sherbini , Taha , Kamrul	100%	10/21/25	10/24/25
Designing Lift Dynamics	Mohammad Kamrul, Taha , El Sherbini , Aqle	100%	10/7/25	10/22/25
Closure				
Further Enhancements	Abdalla Aqle , El Sherbini , Taha , Kamrul	100%	11/3/25	11/5/25
3D Printing	Mohamed El Sherbini , Aqle , Taha , Kamrul	100%	11/9/25	11/23/25
Real Word Assembly	Abdalla Aqle , El Sherbini , Taha , Kamrul	100%	11/9/25	11/23/25
Determining Error	Mohammed Taha , Aqle, Kamrul , El Sherbini	100%	11/11/25	11/16/25
Real Word Testing	Abdalla Aqle , El Sherbini , Taha , Kamrul	100%	11/16/25	11/23/25
Prepare Report	Abdalla Aqle , El Sherbini , Taha , Kamrul	100%	11/26/25	11/30/25
Prepare Progress Presentatior	Mohamed El Sherbini , Aqle , Taha , Kamrul	100%	11/8/25	11/12/25
Submittion of Presentation	Abdalla Aqle , El Sherbini , Taha , Kamrul	100%	11/30/25	11/30/25

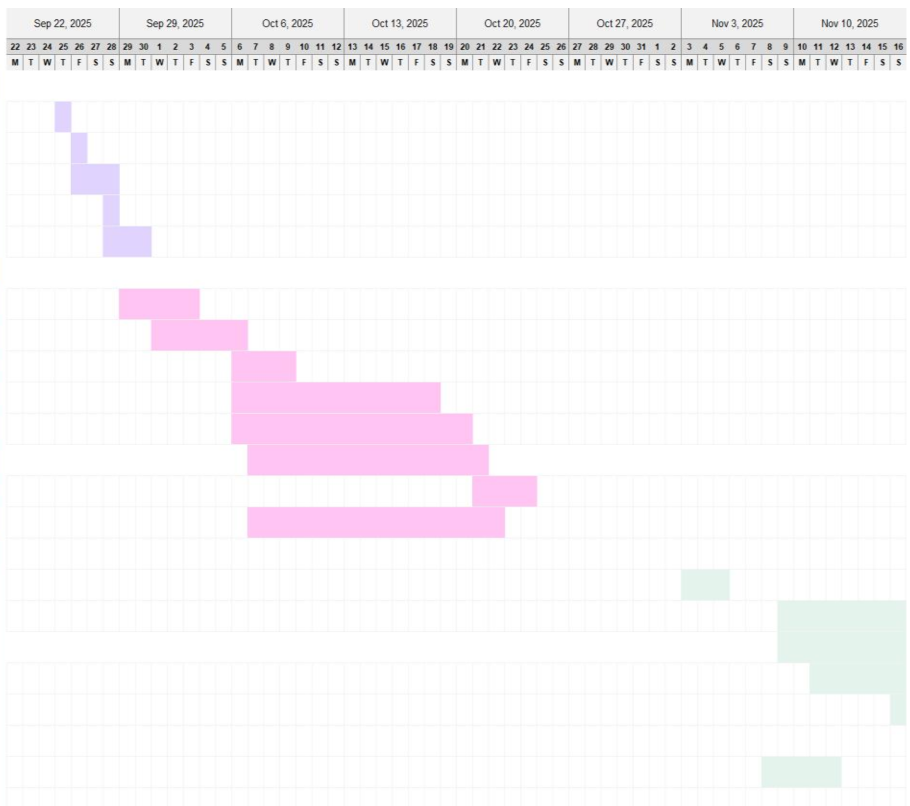

Figure 1: Gantt Chart

1. Project Formulation

Note: Each links combinations result in different trace path graphs made by the two hooks transferring the object from the ground to the higher level.

2.1 Overview

This project focuses on designing a mechanical lifting system driven by a hand-operated crank and inspired by the motion seen in the reference video. The mechanism uses a series of connected links that alternate in lifting an object during different phases of the crank rotation. A gear was added to meet the requirement of including at least one motion transmission component. The system was first sketched, then modeled in Linkage software, followed by a full 3D CAD assembly in SolidWorks. The final model was used to perform kinematic and dynamic analysis before constructing a physical prototype to verify the motion.

2.2 Design Specifications

The design of the mechanism follows a set of specifications and constraints that ensure proper functionality, manufacturability, and compliance with project requirements. The key design specifications are listed below:

1. **Linkage Design:**
   1. **Determination of number of necessary links:**
      When taking one side of the mechanism it is evident that it's working on a 6 bar linkage mechanism with 3 links fixed to the ground:

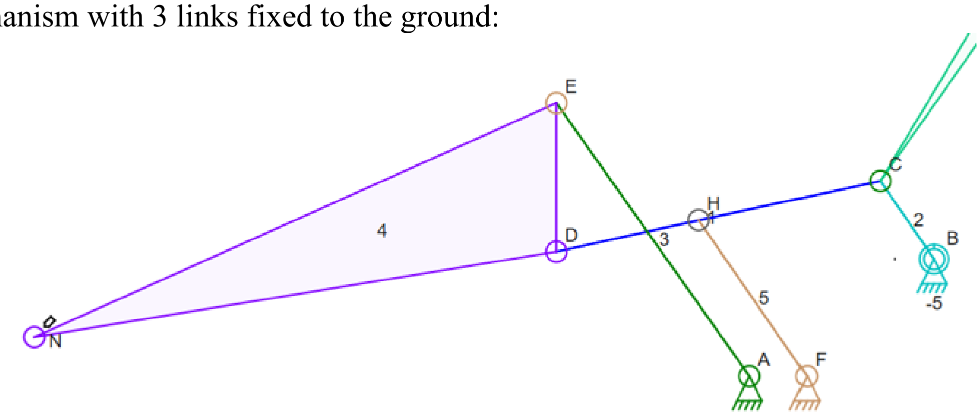

Figure 2: Linkage Design

   2. **Kutzbach Mobility Criterion:**

$$m = 3(n-1) - 2j_1$$
$$m = 3(6-1) - 2(7) = 1$$

      Thus the mechanism is valid for one crank rotation (One Motor).
   3. **Linkage App Simulation:**

Note: Each links combinations result in different trace path graphs made by the two hooks transferring the object from the ground to the higher level.

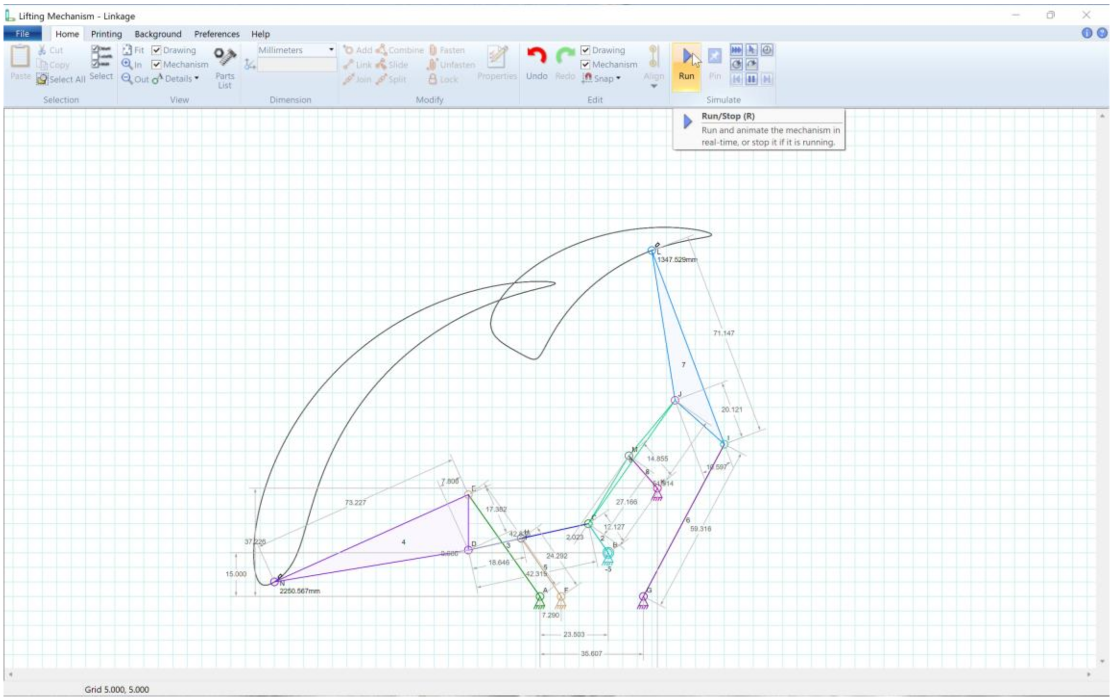

Figure 3: Linkage Final Design with Dimensions

The lengths of the links were then brought to solidworks after checking that the mechanism is working on linkage.

2. **Input Method**
   The machine must be driven by a single hand-operated crank that provides a constant angular velocity **during analysis**.
3. **Motion Transmission Component**
   The mechanism must include at least one additional motion transmission element. Available options include gear, a cam, a belt, or a chain. For this project, a **gear** was selected and integrated into the linkage system.

2.3 Individual parts/component design

1. **Crank:**
   The crank is designed with a circular shape to provide more stability since the center of mass will be in the middle of the link so the center of mass will not be moving throughout the rotation. In addition, the choice of a circular crank is the best in terms of attaching a spur gear to it.

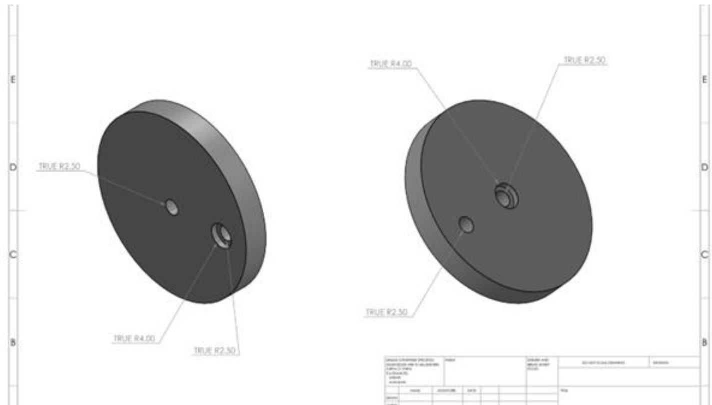

Figure 4: Crank

2. **The Ground:**
   The ground is designed in a way that the columns which are holding the links are filleted from the bottom to prevent failure during operation due to bending and tension. Some columns' fillets are even attached to each other acting as horizontal beams for rigidity. The ground is made to be under the geometric limitations of the project description.

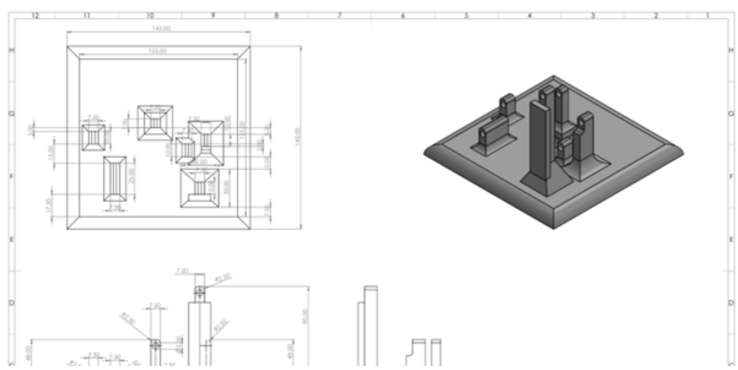

Figure 5: Ground

3. **Bolts:**
   Male and Female attachment of bolts is chosen to be the method of coupling the links together. However, due to the small size of the project specifications outlined to us, we chose to use metal female and male bolts for attachment instead of 3d printing it. Especially, The Bolt in the middle of the mechanism that connects the crank with the two sides of the mechanism, because of the double shear stress applied on it.

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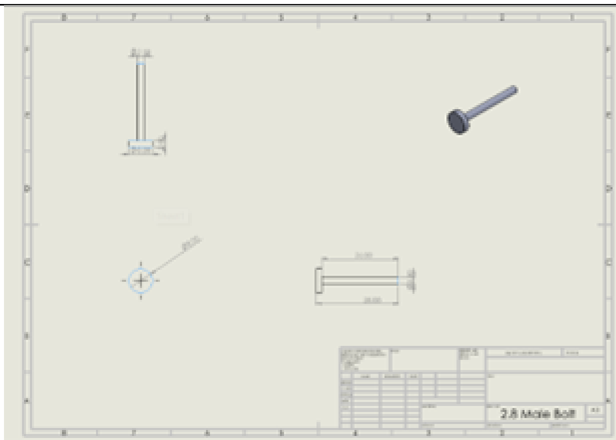

Figure 6: Engineering Drawing of Male Bolt

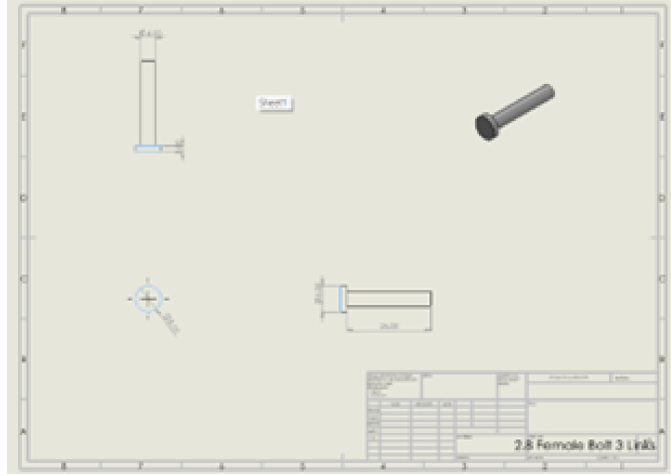

Figure 7: Engineering Drawing of Female Bolt

4. **The First Hook:**
   The design of the hook is unique. The outer profile is crescent shape but with almost 270 degrees angle. This doesn’t allow the object to leave the hook while transferring it to the higher level like a projectile. The inner profile of the second half of the hook is designed with an opposite direction curve to facilitate the movement (rolling) of the object on the hook while in contact with the second hook. Moreover, the link was hollowed to decrease the load on the crank

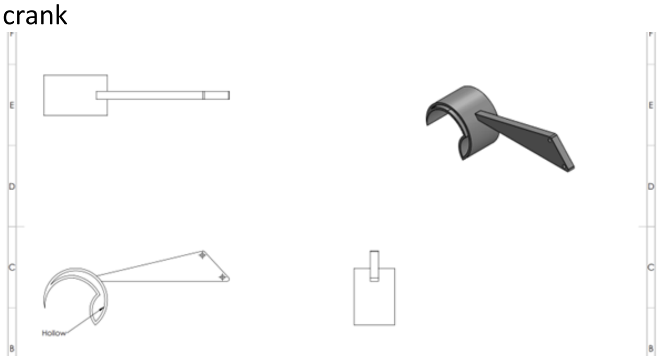

Figure 8: First Hook

5. **The Second Hook:**
   The second hook is designed to have a very open shape to act as a projectile to through the object away to the aimed level.

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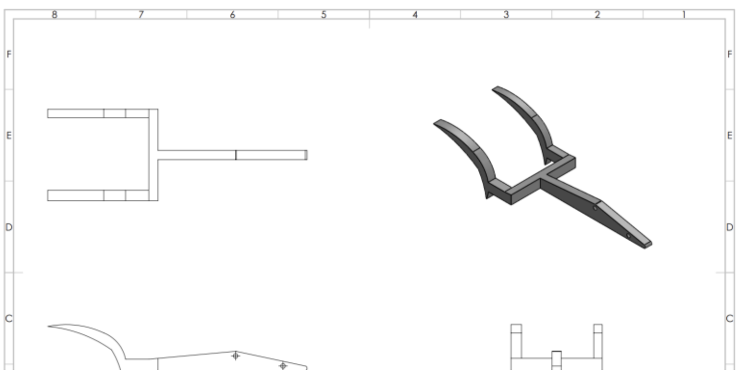

Figure 9: Second Hook

2.4 Final Design

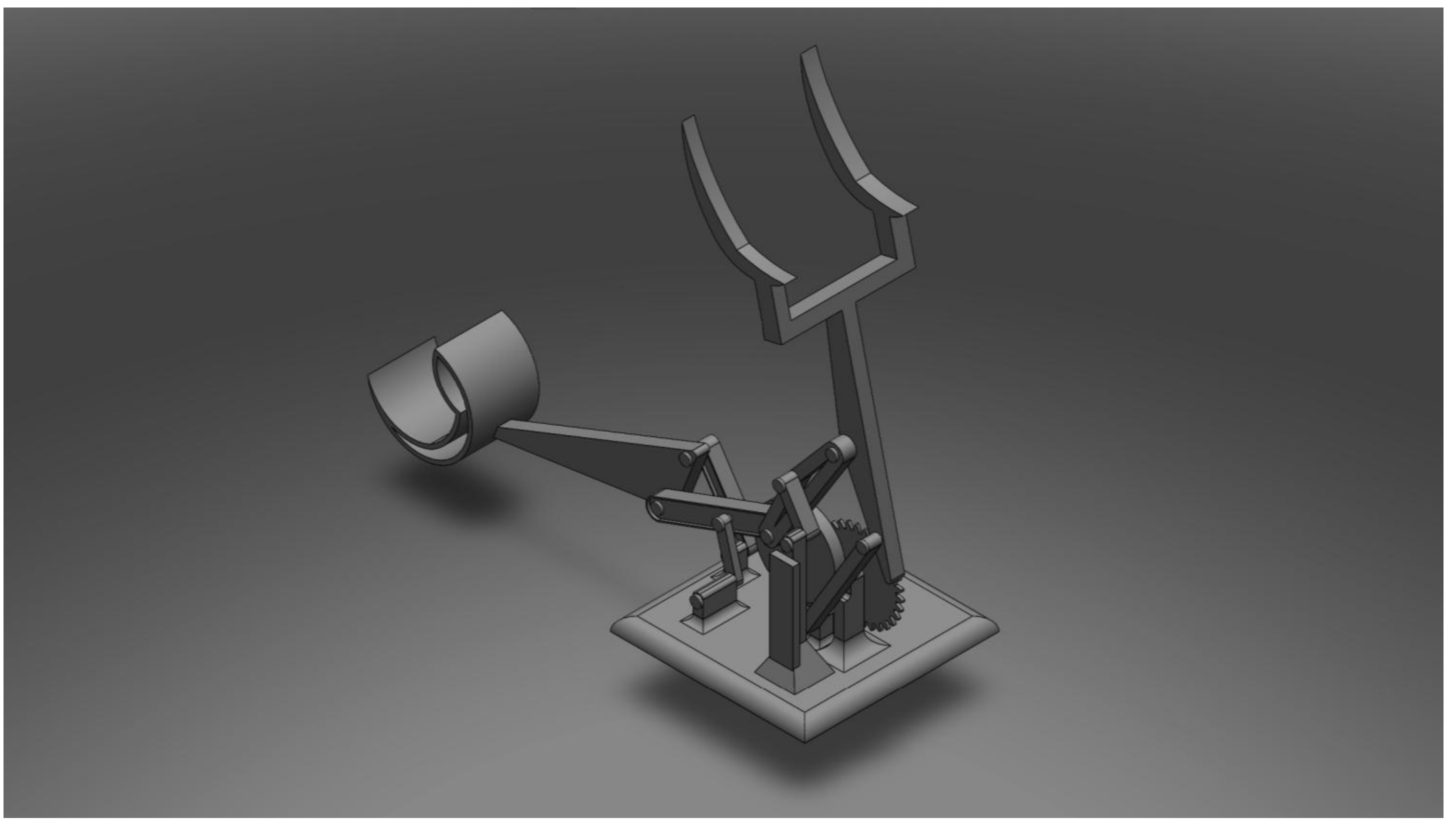

Figure 10: Final 3D Model in SolidWorks

The final design consists of a hand-cranked lifting mechanism built from a set of rigid links arranged to replicate the alternating lifting motion shown in the reference video. A gear was incorporated into the system to control the motion transfer between links and to satisfy the project requirement of using at least one

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transmission element. The completed SolidWorks assembly includes correctly dimensioned parts, defined pivot locations, and a smooth crank-driven motion that allows each side of the mechanism to lift the object during different phases of rotation.

2.5 Discussion

The engineering drawings and final prototype demonstrate that the designed mechanism successfully translates the conceptual motion into a functional lifting system. The CAD drawings confirm proper link dimensions, accurate hole placements, and correct alignment of the gear and crank, all of which contributed to achieving smooth motion during assembly. The prototype closely matches the virtual model and reproduces the alternating lifting action observed in the simulation. Minor deviations in motion were mainly due to real-world factors such as friction, joint clearance, and material flexibility, which are not fully captured in the software environment. Overall, the comparison between the drawings, the SolidWorks model, and the physical build shows strong consistency and validates the design process from initial sketching to final fabrication.

2. Engineering Design and Analysis

3.1 Numerical Analysis: Position, Velocity, Acceleration, and Force

Closed Kinematic Chain – Vector Loop Derivations
1. Position Equations
First Loop:

$$Z_1 + Z_2 - Z_3 - Z_4 = 0$$
$$ae^{j\theta_1} + be^{j\theta_2} - ce^{j\theta_3} - de^{j\theta_4} = 0$$

Separating real and imaginary parts:
Real:

$$a\cos\theta_1 + b\cos\theta_2 - c\cos\theta_3 - d\cos\theta_4 = 0$$

Imaginary:

$$a\sin\theta_1 + b\sin\theta_2 - c\sin\theta_3 - d\sin\theta_4 = 0$$

Second Loop:

$$ee^{j(\theta_4+\pi)} + fe^{j\theta_6} + ge^{j\theta_7} - he^{j\theta_2} - ie^{j\theta_9} = 0$$

Real:

$$e\cos(\theta_4+\pi) - f + g\cos\theta_7 - h\cos\theta_2 - i\cos\theta_9 = 0$$

Imaginary:

$$e\sin(\theta_4+\pi) + g\sin\theta_7 - h\sin\theta_2 - i\sin\theta_9 = 0$$

2. Velocity Equations
For each link:

$$\dot{Z}_i = j\omega_i Z_i$$

First Loop:

$$ja\omega_1 e^{j\theta_1} + jb\omega_2 e^{j\theta_2} - jd\omega_4 e^{j\theta_4} = 0$$

Real:

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$$\omega_1 a\cos\theta_1+\omega_2 b\cos\theta_2-\omega_4 d\cos\theta_4=0$$

Imaginary:

$$-\omega_1 a\sin\theta_1-\omega_2 b\sin\theta_2+\omega_4 d\sin\theta_4=0$$

Second Loop:

$$je\omega_5 e^{j(\theta_4+\pi)}+jg\omega_7 e^{j\theta_7}-jh\omega_2 e^{j\theta_2}-ji\omega_9 e^{j\theta_9}=0$$

Real:

$$\omega_4 e\cos(\theta_4+\pi)+\omega_7 g\cos\theta_7-\omega_2 h\cos\theta_2-\omega_9 i\cos\theta_9=0$$

Imaginary:

$$-\omega_4 e\sin(\theta_4+\pi)-\omega_7 g\sin\theta_7+\omega_2 h\sin\theta_2+\omega_9 i\sin\theta_9=0$$

3. Acceleration of a Single Link

Given:

$$R=ae^{j\theta},\quad V=j\omega ae^{j\theta}$$

$$A=\frac{dV}{dt}=-\omega^2 ae^{j\theta}+j\alpha ae^{j\theta}$$

Expanding:

$$A=a[-\omega^2(\cos\theta+j\sin\theta)]+ja[\alpha(\cos\theta+j\sin\theta)]$$

$$A=a(\alpha\cos\theta-\omega^2\sin\theta)j+a(\alpha\sin\theta+\omega^2\cos\theta)$$

4. Acceleration Equations for Full Mechanism

First Loop:

Imaginary:

$$a\left(\alpha_1\cos\theta_1-\omega_1^2\sin\theta_1\right)\ +\ b\left(\alpha_2\cos\theta_2-\omega_2^2\sin\theta_2\right)-d\left(\alpha_4\cos\theta_4-\omega_4^2\sin\theta_4\right)=0$$

Real:

$$-\,a\left(-\,\alpha_1\sin\theta_1+\omega_1^2\cos\theta_1\right)\ -\ b\left(-\,\alpha_2\sin\theta_2+\omega_2^2\cos\theta_2\right)+d\left(-\,\alpha_4\sin\theta_4+\omega_4^2\cos\theta_4\right)=0$$

Second Loop:

Imaginary:

$$e\left(\alpha_4\cos(\theta_4+\pi)-\omega_4^2\sin(\theta_4+\pi)\right)+g\left(\alpha_7\cos\theta_7-\omega_7^2\sin\theta_7\right)-h\left(\alpha_2\cos\theta_2-\omega_2^2\sin\theta_2\right)$$
$$-\,i\left(\alpha_9\cos\theta_9-\omega_9^2\sin\theta_9\right)=0$$

Real:

$$-e\left(-\,\alpha_5\sin(\theta_4+\pi)+\omega_5^2\cos(\theta_4+\pi)\right)\ -\ g\left(-\,\alpha_7\sin\theta_7+\omega_7^2\cos\theta_7\right)+h\left(-\,\alpha_2\sin\theta_2+\omega_2^2\cos\theta_2\right)$$
$$+\,i\left(-\,\alpha_9\sin\theta_9+\omega_9^2\cos\theta_9\right)=0$$

Force analysis:

Crank:

Equations 1–3

(1)

$$P_{1,10,x}+P_{1,3,x}-P_{1,2,x}=0$$

(2)

$$P_{1,10,y} + P_{1,2,y} + P_{1,3,y} = 0$$

(3)

$$T + r_{1,x}P_{1,2,y} - r_{1,y}P_{1,2,x} + r_{1,x}P_{1,3,y} - r_{1,y}P_{1,3,x} = I_1\alpha_1$$

Link 2:

Equations 4–6

(4)

$$P_{1,2,x} + P_{2,4,x} - P_{2,8,x} = m_2 a_{2,x}$$

(5)

$$-P_{1,2,x} + P_{2,4,x} + P_{2,8,y} = m_2 a_{2,y}$$

(6)

$$d_{1,2,2,x}P_{1,2,y} - d_{1,2,2,y}P_{1,2,x} + d_{2,4,2,x}P_{2,4,y} - d_{2,4,2,y}P_{2,4,x} + d_{2,8,2,x}P_{2,8,y} - d_{2,8,2,y}P_{2,8,x} = I_2\alpha_2$$

Link 3:

Equations 7–9

(7)

$$-P_{1,3,x} + P_{3,5,x} + P_{3,9,x} = m_3 a_{3,x}$$

(8)

$$-P_{1,3,y} + P_{3,5,y} + P_{3,9,y} = m_3 a_{3,y}$$

(9)

$$d_{1,3,3,x}P_{1,3,y} - d_{1,3,3,y}P_{1,3,x} + d_{3,5,3,x}P_{3,5,y} - d_{3,5,3,y}P_{3,5,x} + d_{3,9,3,x}P_{3,9,y} - d_{3,9,3,y}P_{3,9,x} = I_3\alpha_3$$

Link 4:

Equations 10–12

(10)

$$-P_{2,4,x} + P_{4,10,x} = m_4 a_{4,x}$$

(11)

$$-P_{2,4,y} - P_{4,10,y} = m_4 a_{4,y}$$

(12)

$$d_{2,4,4,x}P_{2,4,y} - d_{2,4,4,y}P_{2,4,x} + d_{4,10,4,x}P_{4,10,y} - d_{4,10,4,y}P_{4,10,x} = I_4\alpha_4$$

Link 4:

Equations 13–15

(13)

$$-P_{3,5,x} + P_{5,10,x} = m_5 a_{5,x}$$

(14)

$$-P_{3,5,y} + P_{5,10,y} = m_5 a_{5,y}$$

(15)

$$d_{3,5,5,x}P_{3,5,y} - d_{3,5,5,y}P_{3,5,x} + d_{5,10,5,x}P_{5,10,y} - d_{5,10,5,y}P_{5,10,x} = I_5\alpha_5$$

Link 5:

Equations 16–18

(16)

$$\boldsymbol{P_{6,8,x} + P_{6,10,x} = m_6 a_{6,x}}$$

(17)

$$\boldsymbol{P_{6,8,y} + P_{6,10,y} = m_6 a_{6,y}}$$

(18)

$$\boldsymbol{d_{6,8,6,x} P_{6,8,y} - d_{6,8,6,y} P_{6,8,x} + d_{6,10,6,x} P_{6,10,y} - d_{6,10,6,y} P_{6,10,x} = I_6 \alpha_6}$$

Link 7:

Equations 19–21

(19)

$$\boldsymbol{P_{7,9,x} + P_{7,10,x} = m_7 a_{7,x}}$$

(20)

$$\boldsymbol{P_{7,9,y} + P_{7,10,y} = m_7 a_{7,y}}$$

(21)

$$\boldsymbol{d_{7,9,7,x} P_{7,9,y} - d_{7,9,7,y} P_{7,9,x} + d_{7,10,7,x} P_{7,10,y} - d_{7,10,7,y} P_{7,10,x} = I_7 \alpha_7}$$

Link 8:

Equations 22–24

(22)

$$\boldsymbol{-P_{6,8,x} + P_{2,8,x} = m_8 a_{8,x}}$$

(23)

$$\boldsymbol{-P_{6,8,y} - P_{2,8,y} = m_8 a_{8,y}}$$

(24)

$$\boldsymbol{d_{6,8,8,x} P_{6,8,y} - d_{6,8,8,y} P_{6,8,x} + d_{2,8,8,x} P_{2,8,y} - d_{2,8,8,y} P_{2,8,x} = I_8 \alpha_8}$$

Link 9:

Equations 25–27

(25)

$$\boldsymbol{-P_{7,9,x} - P_{3,9,x} = m_9 a_{9,x}}$$

(26)

$$\boldsymbol{-P_{7,9,y} - P_{3,9,y} = m_9 a_{9,y}}$$

(27)

$$\boldsymbol{d_{7,9,9,x} P_{7,9,y} - d_{7,9,9,y} P_{7,9,x} + d_{3,9,9,x} P_{3,9,y} - d_{3,9,9,y} P_{3,9,x} = I_9 \alpha_9}$$

Gear analysis:

1. The design begins with two known constraints: a **centre distance of 40 mm** between gear shafts and a required **velocity ratio of 3:1**.
2. For spur gears in mesh, the velocity ratio is defined by the radii ratio, so $VR = R_1/R_2 = 3$, which gives the relation $R_2 = R_1/3$.

3. Substituting this relation into the centre-distance equation $R_1 + R_2 = 40$gives $R_1 + R_1/3 = 40$, which simplifies to $R_1 = 30$mm.
4. Using the radius ratio, the second gear radius is obtained as $R_2 = 10$mm.
5. The corresponding pitch diameters are then calculated as $PD_1 = 2R_1 = 60$mm and $PD_2 = 2R_2 = 20$mm.
6. A **diametral pitch of 0.5 mm⁻¹** is selected to standardize tooth dimensions for manufacturability.
7. The number of teeth for each gear is determined using $N = 2R \times DP$, giving **30 teeth for Gear 1** and **10 teeth for Gear 2**.
8. The resulting tooth ratio $30:10$ exactly matches the required **3:1 velocity ratio**, confirming that the design satisfies both kinematic and geometric constraints.

3.2 Analytical Model: Position, Velocity, Acceleration, and Force

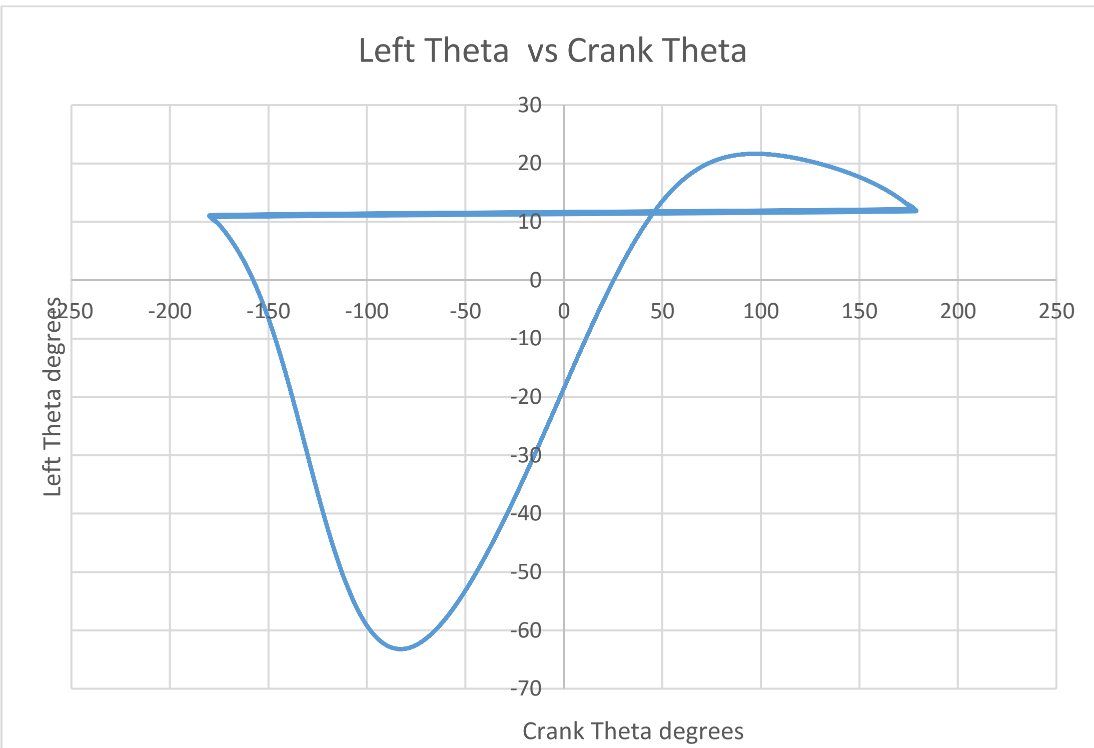


Figure 11: Left Theta vs Crank Theta

Note: Each links combinations result in different trace path graphs made by the two hooks transferring the object from the ground to the higher level.

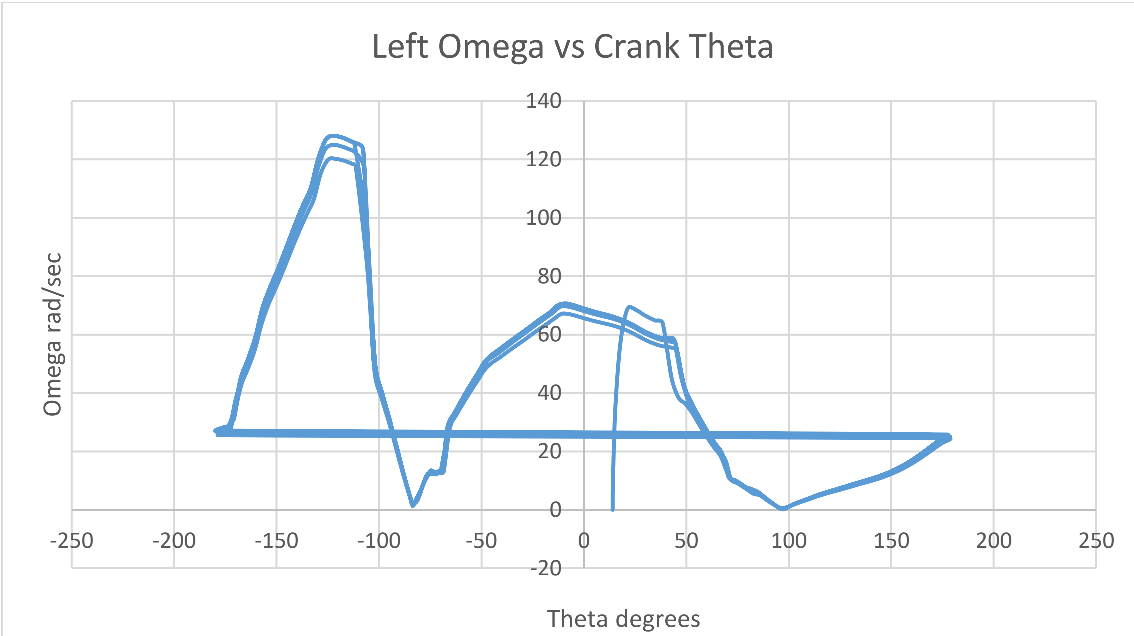


Figure 12:Left Omega vs Crank Theta

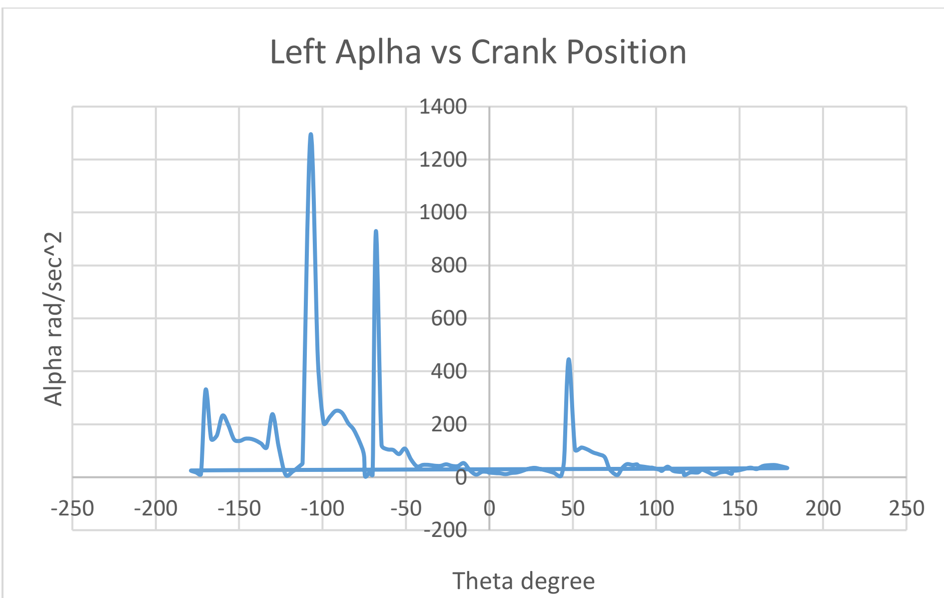


Figure 13 : Left Aplha vs Crank Position

Note: Each links combinations result in different trace path graphs made by the two hooks transferring the object from the ground to the higher level.

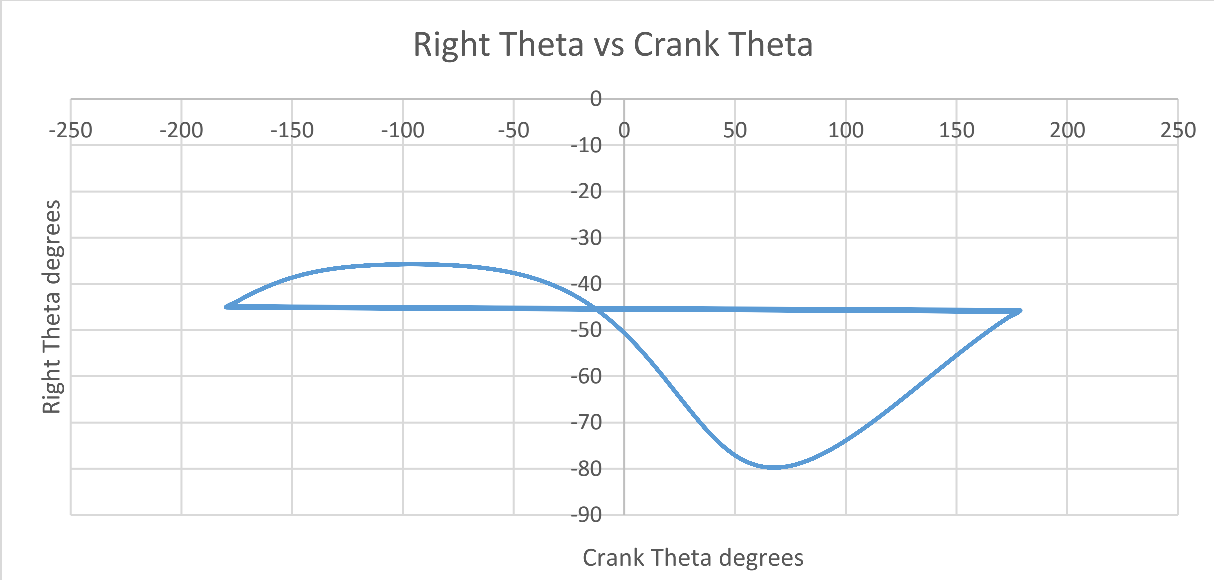


Figure 14: Right Theta vs Crank Theta

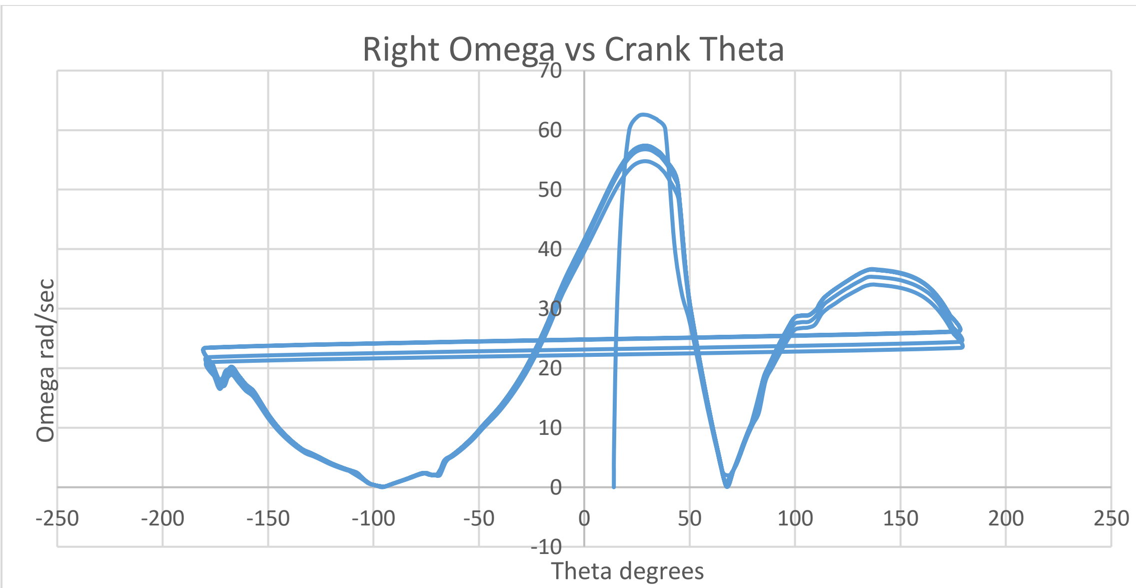


Figure 15: Right omega vs Crank Theta

Note: Each links combinations result in different trace path graphs made by the two hooks transferring the object from the ground to the higher level.

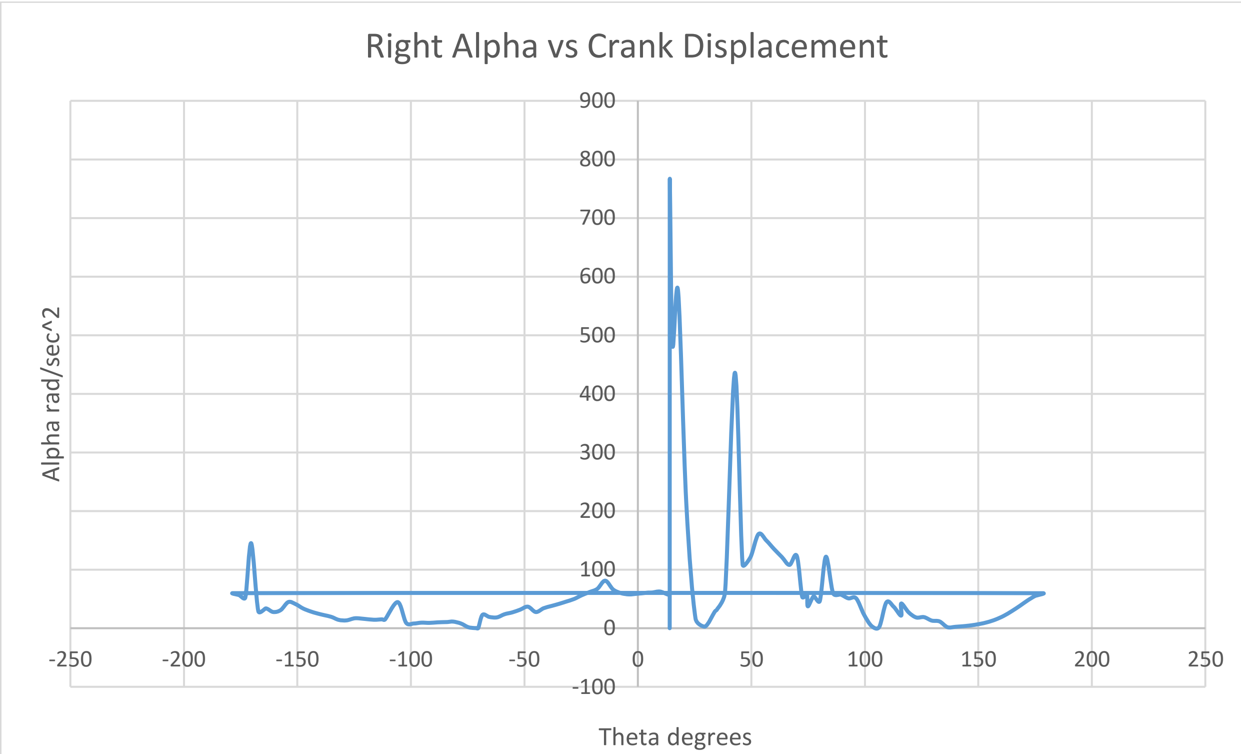


Figure 16: Right Alpha vs Crank Displacement

3.4 Discussion

Comprehensive analysis was performed for the mechanism using both manual calculations and software simulations. Position, velocity, and acceleration plots were generated in SolidWorks Motion and compared with hand calculations and Linkage software outputs to verify the kinematic behavior. For the force analysis, the required force matrix was derived manually as part of the assignment, although solving the matrix was not required. The gears used in the mechanism were selected directly from the SolidWorks Toolbox, and the gear analysis was carried out through hand calculations only to determine the speed ratio and confirm proper engagement. These combined methods ensured that the final design met the expected motion and performance requirements.

Note: Each links combinations result in different trace path graphs made by the two hooks transferring the object from the ground to the higher level.

3. Prototype Construction and Evaluation

4.1 Construction of physical prototype

Figure 17: final prototype

Figure 18: final prototype

4.2 Comparison of Simulation vs Experimental Results

The comparison between the simulation results and the experimental prototype shows that the overall motion, especially the position of the lifting links throughout the rotation, matches closely with the predicted behavior from SolidWorks. The alternating lifting action and general path of the links are consistent in both the virtual and physical models. Minor differences appear due to 3D printing tolerances, slight play in the joints, and friction in the assembled parts, which are not fully represented in the simulation. Despite these small variations, the prototype follows the same positional trends as the simulation, confirming that the design was accurately translated from the digital model to the physical build.

Sources of error of mechanisms 3d prototype:

1. While assembling, we faced a problem of fixing the 3d printed screws. Which we solved by using sandpaper sheets and by gluing them to their correct position. Some other screws were replaced with screws from the workshop.
2. When assembling the hook part which continues the lifting process, the heavy weight resulted in risks regarding failure of parts. This was solved by drilling some symmetrical holes which helps in reducing the weight of the part.

Note: Each links combinations result in different trace path graphs made by the two hooks transferring the object from the ground to the higher level.

3. While assembling the parts, the columns didn't bear the weight of the other parts due to its weak structure and the material used. Therefore the base was reprinted with fillets being applied to all the columns to increase their strength and stability
4. Fillets were applied on all of the columns to stabilize them and prevent them from failure. However, the column for the big sized gear didn't have fillet applied which led to the mechanism not functioning correctly. That column was strapped tightly to a near column to stabilize it which helped in the gear functioning correctly and the whole mechanism functioning as needed.

4.3 Discussion

The final prototype performed very well and successfully reproduced the intended lifting motion observed in the simulation. The mechanism operated smoothly, and the alternating lifting action matched the predicted behavior from the SolidWorks model. During assembly, it was found that the 3D printed bolts lacked sufficient strength and were prone to bending or breaking under load, so they were replaced with steel bolts to ensure proper rigidity and reliable joint connections. This modification improved the stability of the mechanism without affecting its overall function. Apart from minor variations caused by printing tolerances and joint friction, the prototype closely matched the simulated motion, demonstrating that the design was accurately translated from the virtual environment to the physical build.

5. Conclusion

5.1 Conclusion and Discussion

The final prototype of the perpetual lifting mechanism worked perfectly, successfully reproducing the intended alternating lifting motion observed in the SolidWorks simulation. The mechanism operated smoothly throughout the crank rotation, and the kinematic behavior closely matched the predicted position, velocity, and acceleration trends. Minor differences were observed due to 3D printing tolerances, joint friction, and slight play in the assembled links. During assembly, 3D printed bolts were found to be too weak, so steel bolts were used to improve stability and ensure reliable joint connections. Gear selection and analysis were performed by hand, while the gears themselves were obtained from the SolidWorks Toolbox, and they functioned as intended to synchronize motion between the links. Overall, the project successfully demonstrated the full workflow from conceptual sketching and linkage modeling to CAD design, kinematic and dynamic analysis, and prototype fabrication. The close agreement between simulation and experiment validates the design and provides practical insight into the behavior of linkage mechanisms, highlighting the effectiveness of combining theoretical, computational, and hands-on approaches in mechanical system development.

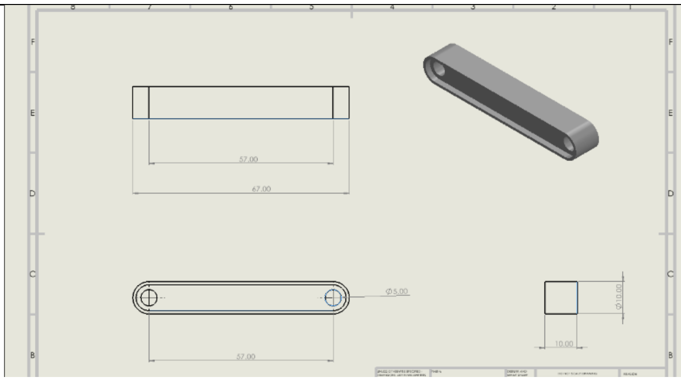
6. References

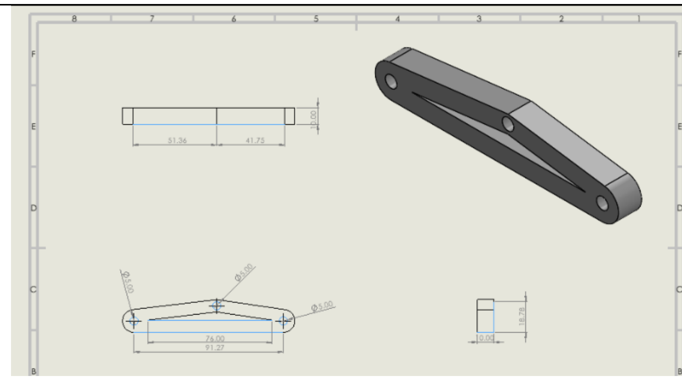
Perpetual lifting machine: https://www.facebook.com/reel/1454269739348994

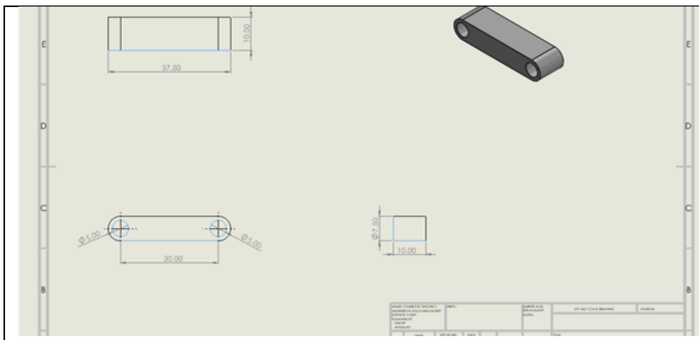
7. Appendices

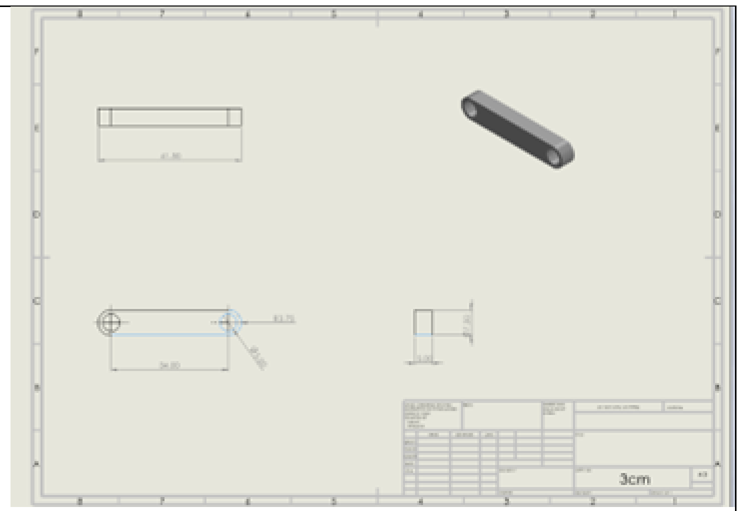
Note: Each links combinations result in different trace path graphs made by the two hooks transferring the object from the ground to the higher level.

Other Links:

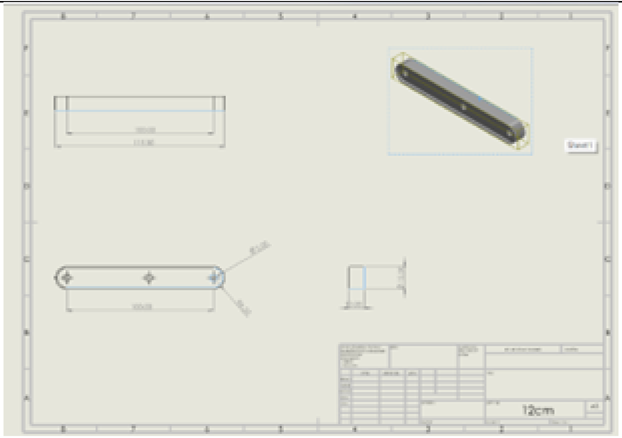


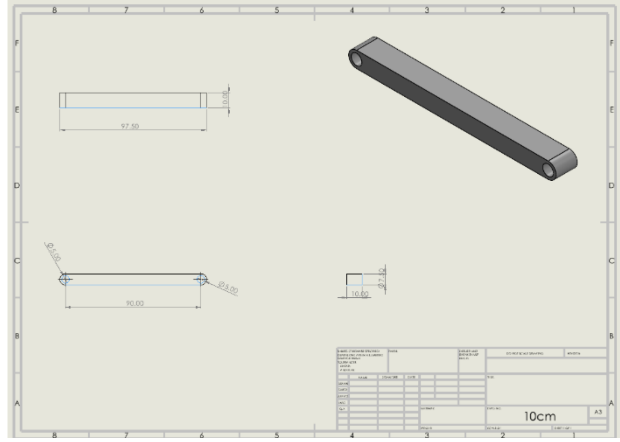






Note: Each links combinations result in different trace path graphs made by the two hooks transferring the object from the ground to the higher level.





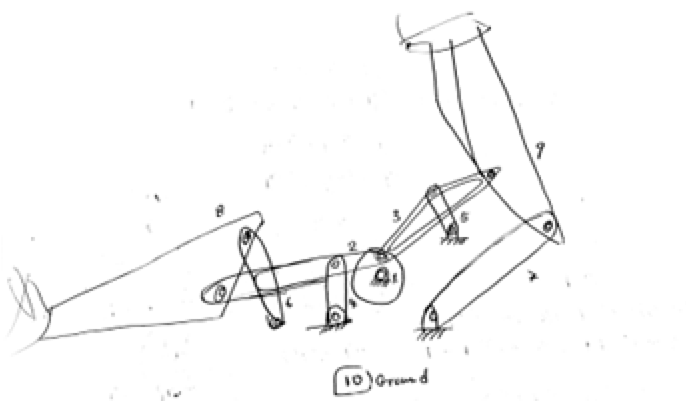
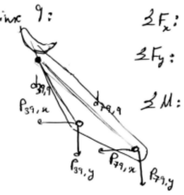


Figure 19: force analysis

Link 9:



$$\Sigma F_x: \; -P_{79,x} - P_{89,x} = m_9 a_{9,x}$$

$$\Sigma F_y: \; -P_{79,y} - P_{89,y} = m_9 a_{9,y}$$

$$\Sigma M: \; (d_{79,9,x} P_{79,y} - d_{79,9,y} P_{79,x}) + (d_{89,9,x} P_{89,y} - d_{89,9,y} P_{89,x}) = I_9 \alpha_9$$

Figure 20: force analysis

Note: Each links combinations result in different trace path graphs made by the two hooks transferring the object from the ground to the higher level.

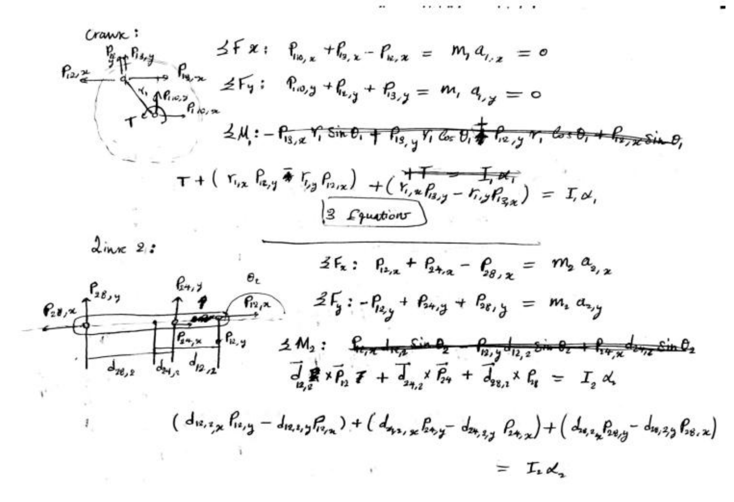


Figure 21: force analysis

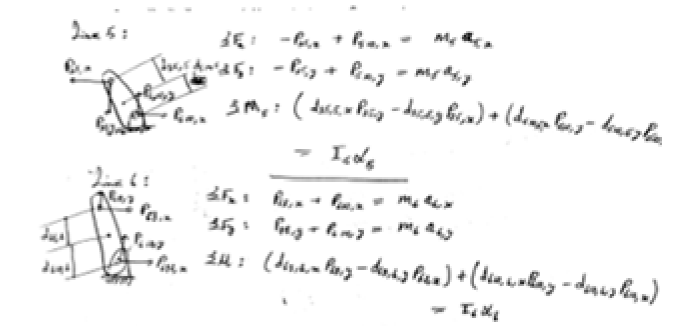


Figure 22: force analysis

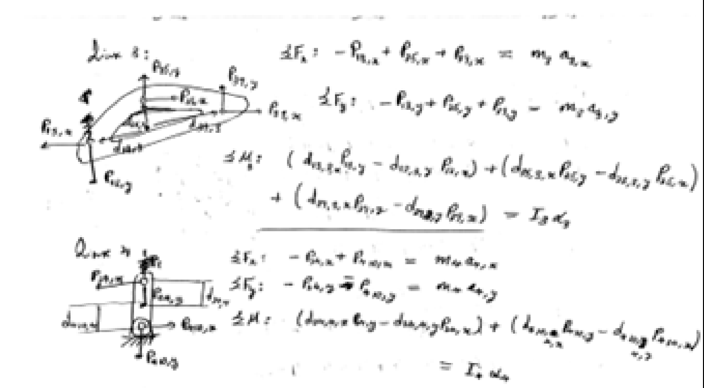


Figure 23: force analysis

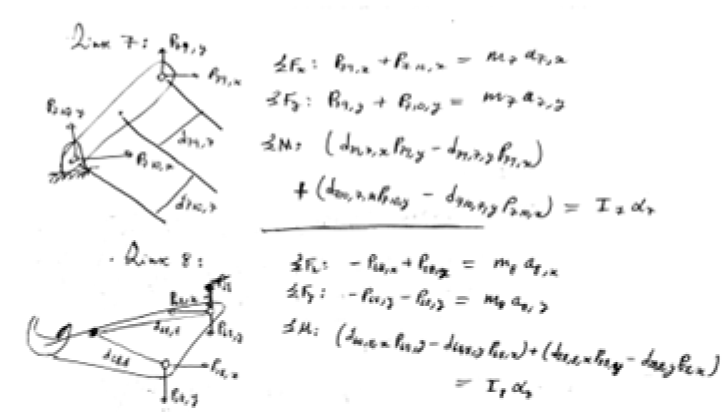


Figure 24: force analysis

Final matrix:

Note: Each links combinations result in different trace path graphs made by the two hooks transferring the object from the ground to the higher level.

$$
\begin{bmatrix}
1 & -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -r_{1,y} & -r_{1,y} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & r_{1,x} & r_{1,x} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
0 & 1 & 0 & 1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -d_{1,2,2,y} & 0 & -d_{2,4,2,y} & -d_{2,8,2,y} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & d_{1,2,2,x} & 0 & d_{2,4,2,x} & d_{2,8,2,x} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -d_{1,3,3,y} & 0 & 0 & -d_{3,5,3,y} & -d_{3,9,3,y} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & d_{1,3,3,x} & 0 & 0 & d_{3,5,3,x} & d_{3,9,3,x} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -d_{2,4,4,y} & 0 & 0 & 0 & -d_{4,10,4,y} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & d_{2,4,4,x} & 0 & 0 & 0 & d_{4,10,4,x} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -d_{3,5,5,y} & 0 & 0 & -d_{5,10,5,y} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & d_{3,5,5,x} & 0 & 0 & d_{5,10,5,x} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -d_{6,8,6,y} & -d_{6,10,6,y} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & d_{6,8,6,x} & d_{6,10,6,x} & 0 & 0 & 0 \\
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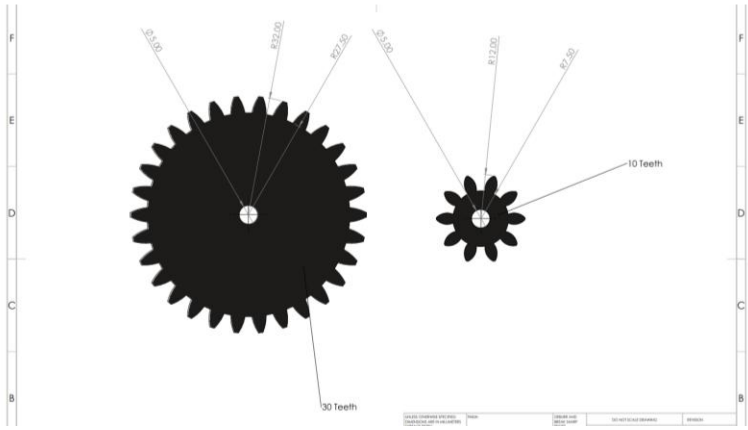
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -d_{2,8,8,y} & 0 & 0 & 0 & 0 & -d_{6,8,8,y} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & d_{2,8,8,x} & 0 & 0 & 0 & 0 & d_{6,8,8,x} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -d_{3,9,9,y} & 0 & 0 & 0 & 0 & -d_{7,9,9,y} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & d_{3,9,9,x} & 0 & 0 & 0 & 0 & d_{7,9,9,x} & 0 & 0
\end{bmatrix}
\times
\begin{bmatrix}
P_{1,10,x} \\ P_{1,2,x} \\ P_{1,3,x} \\ P_{2,4,x} \\ P_{2,8,x} \\ P_{3,5,x} \\ P_{3,9,x} \\ P_{4,10,x} \\ P_{5,10,x} \\ P_{6,8,x} \\ P_{6,10,x} \\ P_{7,9,x} \\ P_{7,10,x} \\ P_{1,10,y} \\ P_{1,2,y} \\ P_{1,3,y} \\ P_{2,4,y} \\ P_{2,8,y} \\ P_{3,5,y} \\ P_{3,9,y} \\ P_{4,10,y} \\ P_{5,10,y} \\ P_{6,8,y} \\ P_{6,10,y} \\ P_{7,9,y} \\ P_{7,10,y} \\ T
\end{bmatrix}
=
\begin{bmatrix}
0 \\ 0 \\ I_1\alpha_1 \\ m_2a_{2,x} \\ m_2a_{2,y} \\ I_2\alpha_2 \\ m_3a_{3,x} \\ m_3a_{3,y} \\ I_3\alpha_3 \\ m_4a_{4,x} \\ m_4a_{4,y} \\ I_4\alpha_4 \\ m_5a_{5,x} \\ m_5a_{5,y} \\ I_5\alpha_5 \\ m_6a_{6,x} \\ m_6a_{6,y} \\ I_6\alpha_6 \\ m_7a_{7,x} \\ m_7a_{7,y} \\ I_7\alpha_7 \\ m_8a_{8,x} \\ m_8a_{8,y} \\ I_8\alpha_8 \\ m_9a_{9,x} \\ m_9a_{9,y} \\ I_9\alpha_9
\end{bmatrix}
$$

Figure 25: Force analysis matrix



Note: Each links combinations result in different trace path graphs made by the two hooks transferring the object from the ground to the higher level.

Figure 26: final prototype

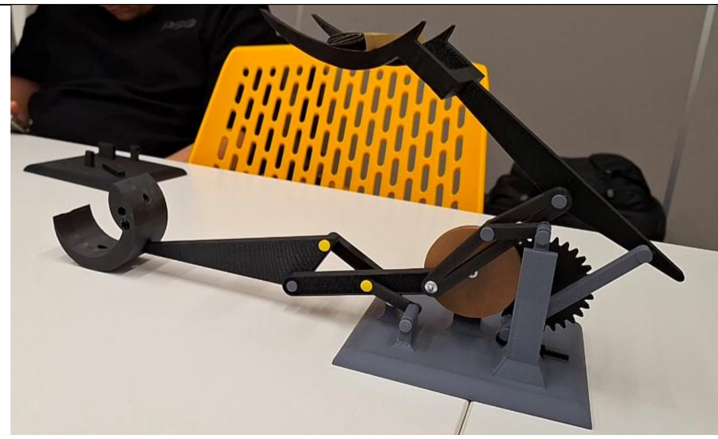


Figure 27: final prototype

We are given a distance : 40 mm
Between centers.

Velocity Ratio is $\boxed{3:1}$

(1) $R_1 + R_2 = 40$, $\frac{\omega_1}{\omega_2} = \frac{R_2}{R_1}$

$\boxed{R_2 = \frac{R_1}{3}}$ $\frac{R_2}{R_1} = \frac{1}{3}$

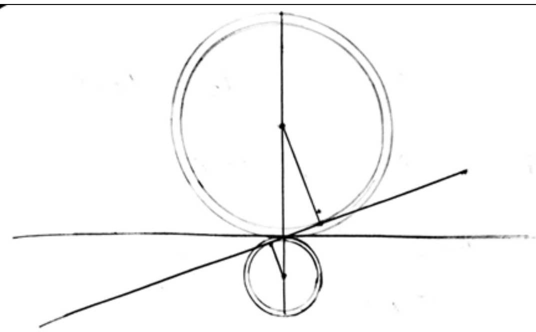
$\longrightarrow R_1 + \frac{R_1}{3} = \frac{4}{3}R_1 = 40$,

$\boxed{R_1 = \frac{3 \times 40}{4} = 30\ mm}$

So $\boxed{R_2 = 10\ mm}$

$PD_1 = 60\ mm$
$PD_2 = 20\ mm$

Figure 28: gear analysis



Diametral pitch = $\frac{1}{2}\ mm^{-1}$

$2 \times R \times DP = N$

For First Gear $\longrightarrow$ $N = 2 \times 30 \times \frac{1}{2} = 30$ teeth

For second Gear $\longrightarrow$ $N = 2 \times 10 \times \frac{1}{2} = 10$ teeth

Figure 29: gear analysis

Note: Each links combinations result in different trace path graphs made by the two hooks transferring the object from the ground to the higher level.